

Aadhav Sivakumar

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EDUCATION

New York University, Tandon School of Engineering

Master of Science in Mechatronics and Robotics | GPA 3.95

Brooklyn, NY

Aug 2024 – May 2026

- **Relevant Coursework** | Reinforcement Learning & Optimal Control (MPC, iLQR/DDP, Q-learning, actor-critic)

University of California, Santa Cruz

Bachelor of Science in Robotics Engineering, Minor in Electrical Engineering | GPA 3.8

Santa Cruz, CA

Jun 2020 – Jun 2024

PROFESSIONAL EXPERIENCE

AI Engineer

Jan 2026 – Sep 2026

Roboflow | Edge AI Engineering Team

New York, NY

- Evaluated NVIDIA's Cosmos 3 omnimodal world model family and benchmarked Cosmos 3 Edge on Jetson Thor.
- Worked with NVIDIA to apply synthetic data generation with PAIDF AnomalyGen to improve vision model performance from few real defect examples; found training iterations, not dataset size, drive the gains.
- Trained and deployed detection, instance segmentation, and keypoint models (RF-DETR, RF-DETR-Seg, YOLO) with neural architecture search, dataset augmentation, and TensorRT/ONNX inference on multi-camera Jetsons.
- Technical DRI for the Roboflow x Standard Bots partnership: built the active-learning loop feeding robot-captured imagery back into training sets and root-caused offline inference failures on robots.
- Designed a custom Docker container for robot control, driving an arm directly from user-configurable Roboflow Workflows as a sidecar to the inference stack; demoed as closed-loop visual servoing on an xArm 5.

Graduate Robotics Researcher

Jan 2026 – Jun 2026

LAIR, NYU Tandon School of Engineering | Advised by Professor Christopher Clark

Brooklyn, NY

- Developed an Isaac Sim Real2Sim pipeline turning 3D Gaussian Splatting scans into digital twins, generating 1k+ randomized scripted demonstrations at 100% in-sim success for sim-to-real augmentation.
- Fine-tuned π_0 and $\pi_{0.5}$ flow-matching VLA policies (openpi/JAX) on bimanual Ultra robot teleoperation data, benchmarked against an ACT baseline.
- Built a bimanual SO-ARM101 LeRobot teleoperation and data-collection stack on a Jetson Orin Nano.
- Assembled and deployed a BlueROV2 for 3D reconstruction of underwater shipwrecks.

Graduate Teaching Assistant

Sep 2025 – May 2026

NYU Tandon School of Engineering | Foundations of Robotics, Math for Robotics, Mechatronics

Brooklyn, NY

- Built a Python UR10e MuJoCo simulation as the final project to assess forward/inverse kinematics and dynamics.
- Taught Jacobian-based IK, state-space control, and motion planning to 80+ graduate students in weekly sessions.
- Wrote the Mechatronics midterm and STM32 NUCLEO-H503RB reference code (USART, ADC, servos, sensors).

Field Robot Technician

Aug 2025 – Jan 2026

Starship Technologies | Fordham University Hub

Bronx, NY

- Root-cause analysis on fleet faults; repaired boards, sensors, and power systems; camera/LiDAR calibration.

Undergraduate Research Assistant

Jun 2023 – Jun 2024

UCSC Tactile Manipulation Lab | Professor Tae Myung Huh

Santa Cruz, CA

- Developed soft robotic CapSense tactile sensors on flexible PCBs, enabling slip-based reorientation at 78% success.

PROJECTS

Unitree Go2 RL Locomotion | Isaac Lab, RSL-RL, Sturm on NYU HPC

Nov 2025 – Dec 2025

- Trained a PPO locomotion policy with a custom PD controller, Raibert gait heuristic, and reward shaping.

Glass-2-Bot Mobile Manipulator | Google Glass, Raspberry Pi, Arduino, OpenCV

Apr 2025 – May 2025

- Object detection on Google Glass video streamed to a Raspberry Pi drives an Arduino grasping state machine.

SMART Compost Sorting | OpenCV, DexNet, Orbec Astra, Franka Panda

Sep 2023 – May 2024

- Senior capstone using DexNet grasping with depth-camera-to-robot calibration for autonomous compost sorting.

TECHNICAL SKILLS

Languages & Tools | Python, C/C++, CUDA, Bash, MATLAB, Linux, Git, Docker, TensorRT, ROS 2, xArm SDK

Robot Learning | PyTorch, JAX, LeRobot, $\pi_0/\pi_{0.5}$, ACT, action chunking, VLA fine-tuning, imitation learning, RL

Data & Evaluation | dataset curation, annotation, active learning, data-scaling and OOD evals, failure analysis

Simulation & Perception | Isaac Sim/Lab, MuJoCo, Gaussian Splatting, Cosmos, sim-to-real, hand-eye calibration